

## Assignment 8

### Extra problem

$$1- \\ 1. \quad C = \begin{bmatrix} 1 & 0.9 & 0.9 \\ 0.9 & 1 & -0.9 \\ 0.9 & -0.9 & 1 \end{bmatrix}$$

For  $C$  to be positive semi-definite, we need  $\det(C) \geq 0$ .

Let's compute  $\det(C)$ :

$$\begin{aligned} \det(C) &= 1 \times (1 \times 1 - (-0.9 \times -0.9)) - 0.9 (0.9 \times 1 - (0.9 \times -0.9)) \\ &\quad + 0.9 ((0.9 \times -0.9) - (1 \times 0.9)) \end{aligned}$$

$$= (1 - 0.81) - 0.9 (0.9 + 0.81) + 0.9 (-0.81 - 0.9)$$

$$= 0.19 - 1.539 - 1.539$$

$$= \underline{-2.888}$$

$\det(C) < 0$   $C$  is not positive semi-definite.

Another inconsistency, is the value of the correlation coefficient:  $\rho_{12} = 0.9$ ,  $\rho_{13} = 0.9$  while  $\rho_{23} = -0.9$ . It says 1 and 2 are almost identical (high correlation) and 1, 3 are also almost identical. Therefore, 2, 3 should be also almost identical by transitivity. This is not the case here.



## 2 - Correlation matrix generation.

$$P(i, j; \eta, \lambda) = \eta + (1 - \eta) \exp \{ -\lambda |T_i - T_j| \}, \quad \forall i, j, \eta, \lambda \geq 0$$

$P_{ij}$  is semi-definite matrix?

We have:

$$\begin{aligned} \sum_{i,j} \alpha_i \alpha_j P_{ij} &= \underbrace{\eta \sum_{i,j} \alpha_i \alpha_j}_{\geq 0} + (1 - \eta) \sum_{i,j} \alpha_i \alpha_j e^{-\lambda |T_i - T_j|} \\ &= \underbrace{\eta}_{\geq 0} \underbrace{\left( \sum_i \alpha_i \right)^2}_{\geq 0} \end{aligned}$$

Also:

$$e^{-\lambda |T_i - T_j|} = \frac{2\lambda}{\pi} \int_0^\infty \frac{\cos(\omega (T_i - T_j))}{\lambda^2 + \omega^2} d\omega$$

we substitute:

$$\begin{aligned} \sum_{i,j} \alpha_i \alpha_j e^{-\lambda |T_i - T_j|} &= \sum_{i,j} \alpha_i \alpha_j \underbrace{\frac{2\lambda}{\pi}}_{\geq 0} \underbrace{\int_0^\infty \frac{1}{\lambda^2 + \omega^2}}_{\geq 0} \underbrace{\sum_{i,j} \alpha_i \alpha_j \cos(\omega (T_i - T_j)) d\omega}_{\geq 0 \text{ squared modulus}} \\ &= \left| \sum_i \alpha_i e^{i\omega T_i} \right|^2 \end{aligned}$$

Therefore  $P_{ij}$  is positive semi-definite since  $\lambda \geq 0, \eta \geq 0, (1 - \eta) \geq 0$

$$\Rightarrow \sum_{i,j} \alpha_i \alpha_j P_{ij} \geq 0 \quad \forall (\alpha_i).$$